

Customer Shopping Behavior Analysis

1. Project Overview

The project analyzes customer shopping behavior using transactional data from 3900 purchases across various product categories. The goal is to uncover insight into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Rows: 3900

Columns: 18 Key

Features: Customer

Demographics (Age, Gender, Location, Subscription Status) **Purchase Details**
(item Purchased, Category, Size, Color, Purchase Amount, Season)

Shopping Behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shopping Type)

Missing Values: 37 values in review rating column.

3. Exploratory Data Analysis (python)

- Data Loading: Imported the data set using **pandas**.
- Basic Info: Used **df.size**, **df.shape**, **df.info()**, **df.describe(include = 'all')** to check the structure and summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- Missing Data Handling: Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- Column Standardization: Renamed column to snake case for better readability.
- Feature Engineering:
 - ✓ Created **age groups** column by binning customer ages.
- Dropped Column: **promo code used** (verified if **discount applied** and **promo code used** are same)
- Database integration: Connected python script to MySQL and loaded the cleaned dataframe into the database for SQL analysis.

4. Data Analysis Using SQL

(Answered some business questions)

Q1. Revenue by Gender:

	gender	revenue
►	Male	157890
	Female	75191

Q2. High-Spent Discount User: identified customers who used the discount but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67

Q3. Top 5 Products by Rating: Finding top 5 products with the highest average rating.

	item_purchased	highest_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.79

Q4. Shipping Type Comparison: Compared the average purchase amounts between Standard and Express Shipping.

	shipping_type	average_purchase_amount
▶	Express	60.48
	Standard	58.46

Q5. Subscribers vs. Non-Subscribers: Compared average spent and total revenue across subscription status.

	subscription_status	avg_spend	total_revenue
▶	Yes	59.4919	62645
	No	59.8651	170436

Q6. Customer Segmentation: Classified customers into new, loyal, and returning based on purchase history.

	customer_segment	customer_count
►	Loyal	3116
	Returning	701
	New	83

Q7. Top 3 Products Per Category: most purchased products from the category.

	item_rank	category	item_purchased	total_orders
►	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Q8. Repeat Buyers & Subscribers: Customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
►	Yes	958

Q9. Total Revenue by Age Groups: Calculated total revenue contribution by each age group.

	age_groups	revenue
►	middle aged	89445
	adult	65842
	senior	43164
	young adult	34630

5. Dashboard In Power Bi



6. Business Recommendations

- Boost Subscriptions: Promote exclusive benefits for subscribers.
- Customer Loyalty Programs: Reward repeat buyers to move them into the 'Loyal Segment'.
- Product positioning: Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing: Focus on high-revenue age groups and express shipping users.